

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed OptiCrop system uses machine learning to recommend the most suitable crop based on soil nutrients and weather conditions. It helps farmers make informed decisions, improve crop yield, reduce resource wastage, and increase agricultural productivity through accurate and data-driven recommendations.

Project Overview	
Objective	Develop an AI-based crop recommendation system that predicts the most suitable crop using soil and weather conditions to help farmers improve productivity.
Scope	The system analyzes soil nutrients (N, P, K), temperature, humidity, pH, and rainfall to recommend the best crop through a simple web application.
Problem Statement	
Description	Farmers often struggle to select the right crop due to changing weather conditions and lack of data-driven recommendations, leading to reduced productivity.
Impact	Wrong crop selection can reduce crop yield, waste resources, and decrease farmers' income.
Proposed Solution	

Approach	Build a machine learning model that analyzes soil and environmental data and recommends the most suitable crop.
Key Features	• AI-based crop recommendation • Soil and weather analysis • Fast and accurate predictions • Easy-to-use web interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel i5 or above
Memory	RAM	8 GB
Storage	Disk space	20 GB Free
Software		
Frameworks	Backend Framework	Flask
Libraries	ML Libraries	Scikit-learn, Pandas, NumPy, Matplotlib
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Crop Recommendation Dataset	Kaggle Crop Recommendation Dataset (.csv)